Tarkan Zarrouk

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SUMMARY

Mathematics student at Carleton University with a strong interest in pure mathematics, particularly mathematical analysis, abstract algebra, and rigorous proof-based reasoning. Experienced in frontend development, database integration, and software engineering through startup work and personal projects. Proficient in React, TypeScript, JavaScript, and Python, with an interest in exploring the connections between mathematical structures, computational logic, and programming.

EXPERIENCE

Frontend Web Developer

Kaie **February 2025 - April 2025, Mississauga, Ontario, Canada**

- Led the integration of real-time collaborative editing functionality, enabling concurrent content updates by multiple users and streamlining team workflows.
- Developed and shipped user profile management features and an interactive Explore page, translating product requirements into responsive, user-facing interfaces.
- Built reusable frontend components and refined UI/UX workflows to improve usability, interface consistency, and code maintainability.
- Migrated 2,000+ user records to a new database, helping preserve data integrity and ensure a smooth transition between database systems.
- Collaborated cross-functionally in a startup environment to develop, test, debug, and deploy features while adapting to evolving requirements and tight deadlines.

Code Sensei

Code Ninjas **July 2024 - August 2024, Cooksville, Mississauga, Ontario, Canada**

- Instructed students aged 6–13 in programming fundamentals, adapting lessons to different age groups and skill levels.
- Taught computational thinking, logical reasoning, and debugging through hands-on coding activities and practical exercises.
- Managed classroom environments of up to 30 students and instructors, maintaining engagement and supporting collaborative learning.
- Collaborated with 2–4 colleagues to coordinate lessons, support students, and deliver effective programming instruction.
- Developed and implemented a curriculum enhancement for advanced students, incorporating game development principles and project-based learning modules.

- Interactive linear algebra visualizer for Carleton University MATH 1104 students
- Visualizes matrices, vectors, linear systems, and transformations in 2D and 3D
- Step-by-step solutions to connect calculations with exam techniques
- Exact fraction arithmetic for accurate results
- Designed to strengthen conceptual understanding through interactive exploration
- Built with React and TypeScript, enabling conceptual understanding through interactive exploration and step-by-step solutions.

- Designed and developed a database-driven social media application using React and TypeScript, building practical experience with component-based architecture, application state, and full-stack development.
- Integrated Firebase services to support data persistence and user-facing features, connecting frontend components with backend infrastructure.
- Implemented CRUD workflows for creating, retrieving, updating, and deleting application records, with data handling and validation to improve consistency and reliability.
- Structured application logic around reusable components and maintainable data flows, strengthening understanding of how frontend interfaces interact with backend services.

PROJECT

Linear Algebra Visualization App

October 2026 - October 2026

- Interactive linear algebra visualizer for Carleton University MATH 1104 students
- Visualizes matrices, vectors, linear systems, and transformations in 2D and 3D
- Step-by-step solutions to connect calculations with exam techniques
- Exact fraction arithmetic for accurate results
- Designed to strengthen conceptual understanding through interactive exploration
- Built with React and TypeScript, enabling conceptual understanding through interactive exploration and step-by-step solutions.

Social Media App

October 2024 - January 2025

- Designed and developed a database-driven social media application using React and TypeScript, building practical experience with component-based architecture, application state, and full-stack development.
- Integrated Firebase services to support data persistence and user-facing features, connecting frontend components with backend infrastructure.
- Implemented CRUD workflows for creating, retrieving, updating, and deleting application records, with data handling and validation to improve consistency and reliability.
- Structured application logic around reusable components and maintainable data flows, strengthening understanding of how frontend interfaces interact with backend services.
- Implemented user authentication and authorization using Firebase Auth, securing application access and personal data.

EDUCATION

Honours Bachelor in Mathematics

Carleton University • Ottawa, Ontario, Canada

- Relevant Coursework: Linear Algebra, Multivariable Calculus, Mathematical Analysis, Abstract Algebra
- Developing a foundation in mathematical reasoning, formal proofs, and analytical problem-solving.
- Exploring the intersection of mathematics and programming, with interests in theoretical mathematics and computational applications.

High School Diploma

Father Michael Goetz Catholic Secondary School • Mississauga, Ontario Canada • 2025 • 3.89

- Graduated with a 3.56/4.00 GPA while participating in technology-focused extracurricular activities.
- Participated in the Robotics Team and Computer Science Club, developing technical and collaborative problem-solving skills.
- Competed in Skills Canada Web Design, Ontario Tech Engineering Robotics, Skills Ontario Robotics, and the Waterloo CSIMC mathematics contest.
- Participated in the Information and Communications Technology (ICT) program, gaining exposure to programming, technology, and applied computing.

SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, C++, HTML5, CSS3
- **Frontend Development:** React, Next.js, Tailwind CSS, Responsive UI Design, Component-Based Architecture
- **Backend & Databases:** Firebase, CRUD Operations, Database Integration, Data Validation
- **Computer Vision & AI:** OpenCV, YOLOv5, MediaPipe, Image Processing, Object Detection
- **Embedded Systems:** Arduino, Ultrasonic Sensors, Motor Control, Hardware-Software Integration
- **Tools & Practices:** Git, GitHub, Debugging, Testing, Troubleshooting, Modular Software Design